Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

Submitted By: Mahir Mavani

Name: Mahir Mavani

Reg. No.: 2411021240003

Date: 22-08-2026

GitHub Link: https://github.com/MahirMavani/BUSINESS-ANALYTICS/tree/main/EXPERIMENT-1

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

The objective of this experiment is to integrate business datasets from multiple sources into Microsoft Power BI and develop an interactive sales analytics dashboard. The AdventureWorks datasets contain information related to customers, products, sales transactions, product categories, product subcategories, and sales territories.

The data is connected to Power BI through MySQL and other available data sources. The integrated data is transformed, modeled, and analyzed to generate meaningful business insights related to sales performance, customers, products, categories, regions, and sales trends.

The original reference also frames the problem around integrating multiple business data sources into Power BI for centralized reporting.

Data Sources Used

Dataset/Table	Purpose	Important Columns
AdventureWorks Customer Lookup	Customer information	CustomerKey, FirstName, LastName, Gender, EducationLevel
AdventureWorks Product Lookup	Product information	ProductKey, ProductName, ProductColor, ProductCost, ProductPrice
AdventureWorks Sales Data 2022	Sales transactions	CustomerKey, ProductKey, OrderDate, OrderQuantity, TerritoryKey
Product Subcategories Lookup	Product classification	ProductSubcategoryKey, ProductCategoryKey, SubcategoryName
Product Categories Lookup	Product classification	ProductCategoryKey, CategoryName
Territory Lookup	Regional information	SalesTerritoryKey, Region, Country, Continent

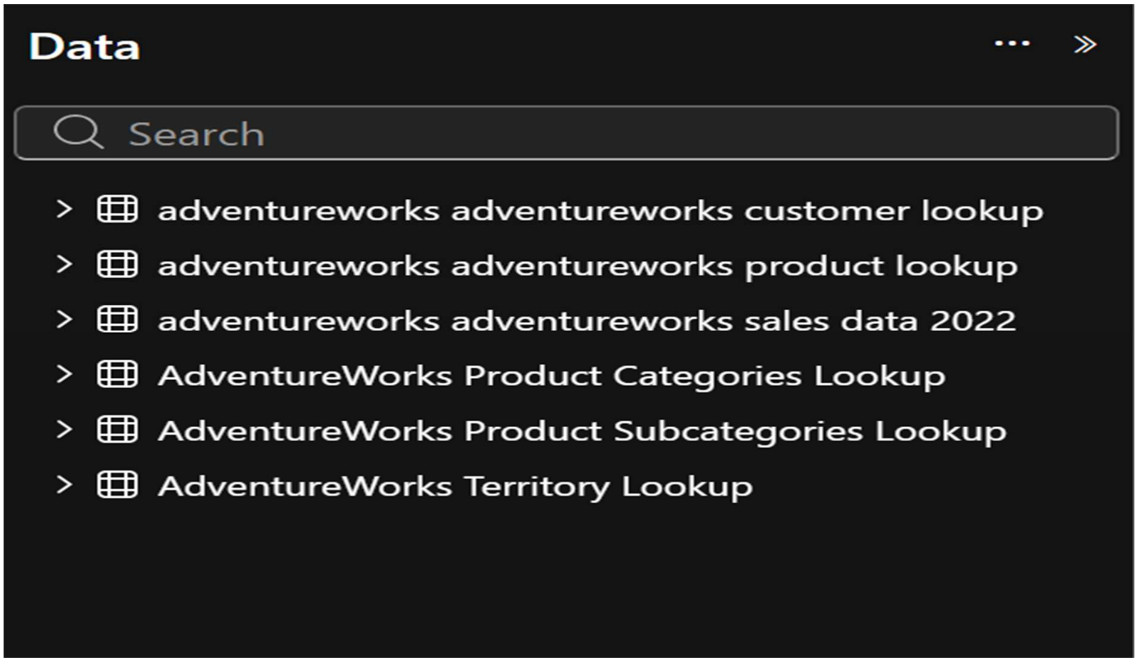

Figure 1: Data sources imported into Power BI

Methodology

➔ Data Collection

- Collected AdventureWorks customer data.
- Collected product and product classification data.
- Collected sales transaction data for 2022.
- Collected sales territory information for regional analysis.
- The datasets were obtained from MySQL and other available data sources.

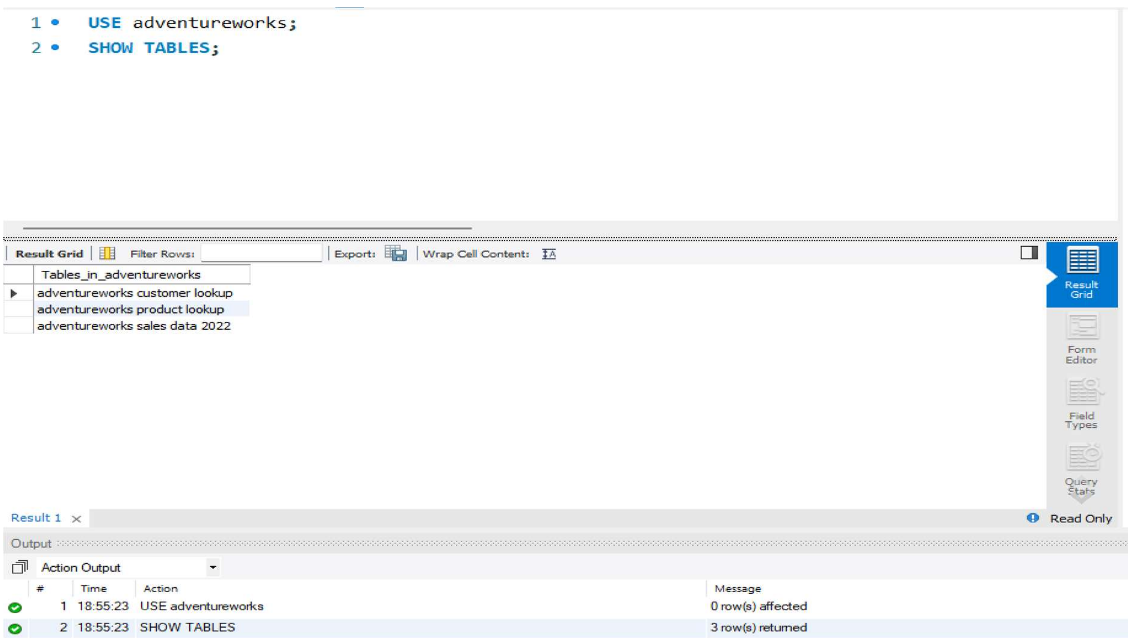

Figure 2: AdventureWorks tables available in MySQL Workbench

➔ Data Import

Power BI Desktop was opened and the required datasets were imported using the Get Data option. MySQL Database was selected as one of the data sources, and the required AdventureWorks tables were loaded into Power BI.

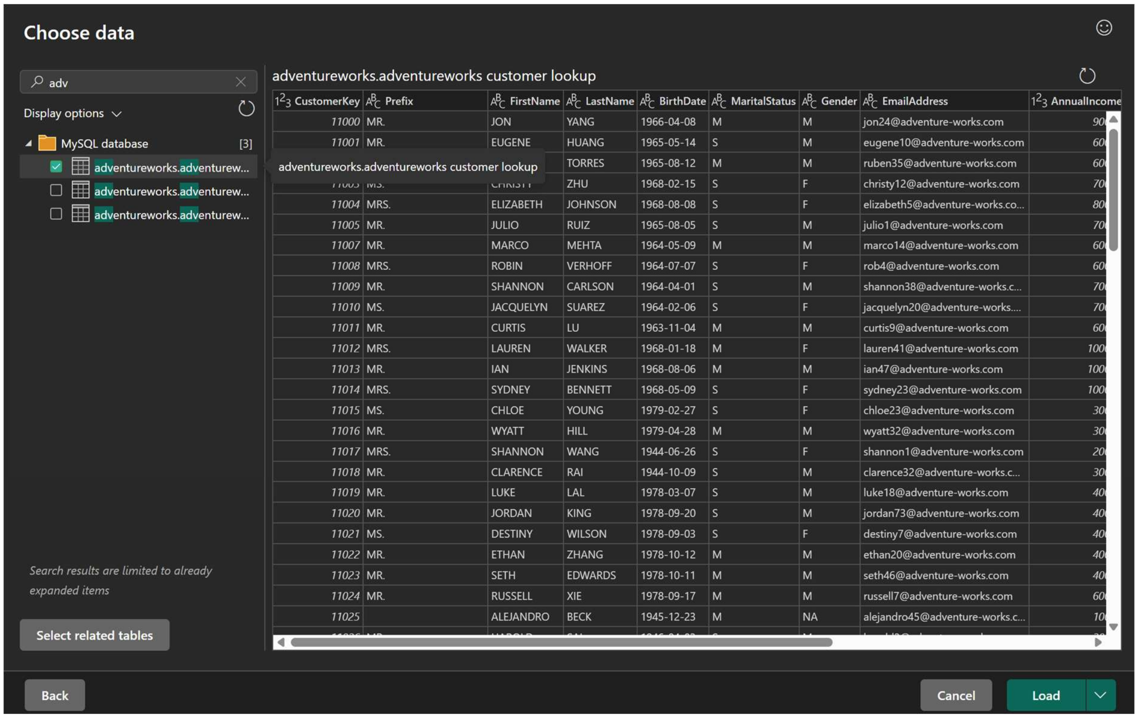

Figure 3: Importing AdventureWorks data from MySQL into Power BI

➔ Data Cleaning and Transformation

The imported datasets were reviewed and transformed using Power Query. Appropriate data types were assigned to numerical, text, and date columns. The data-quality indicators were checked to identify errors and empty values. Unnecessary rows were removed where required, and the datasets were prepared for further data modeling and analysis.

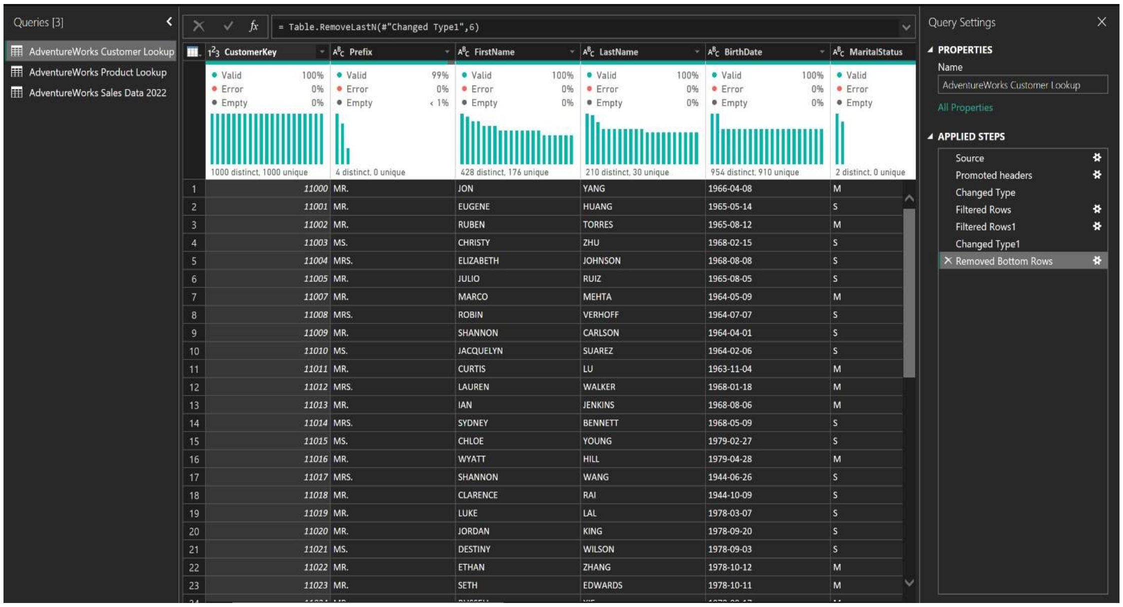

Figure 4(a): Customer data transformation in Power Query

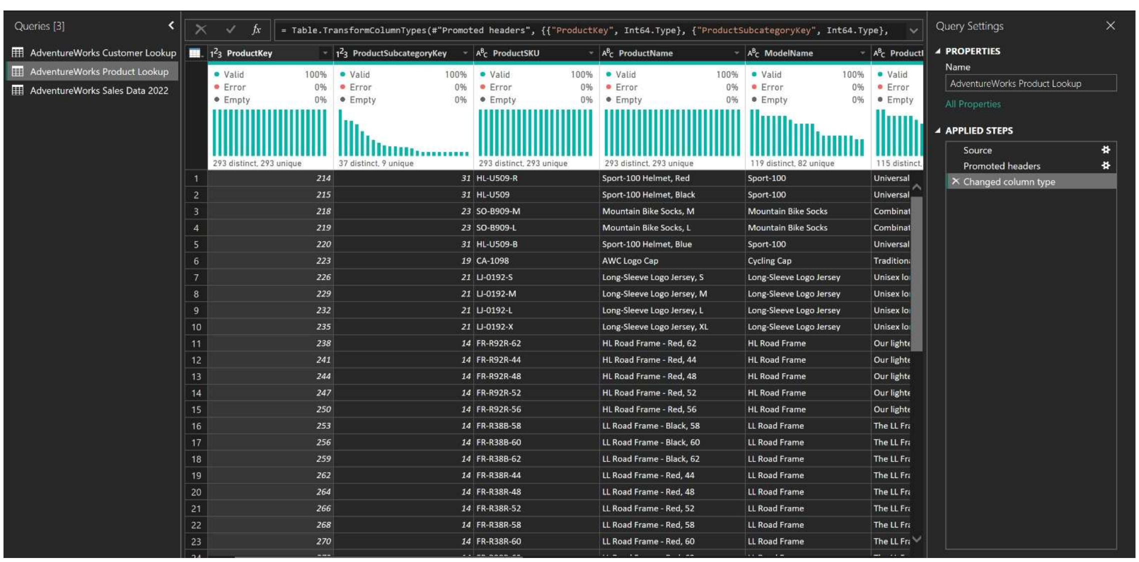

Figure 4(b): Product data transformation in Power Query

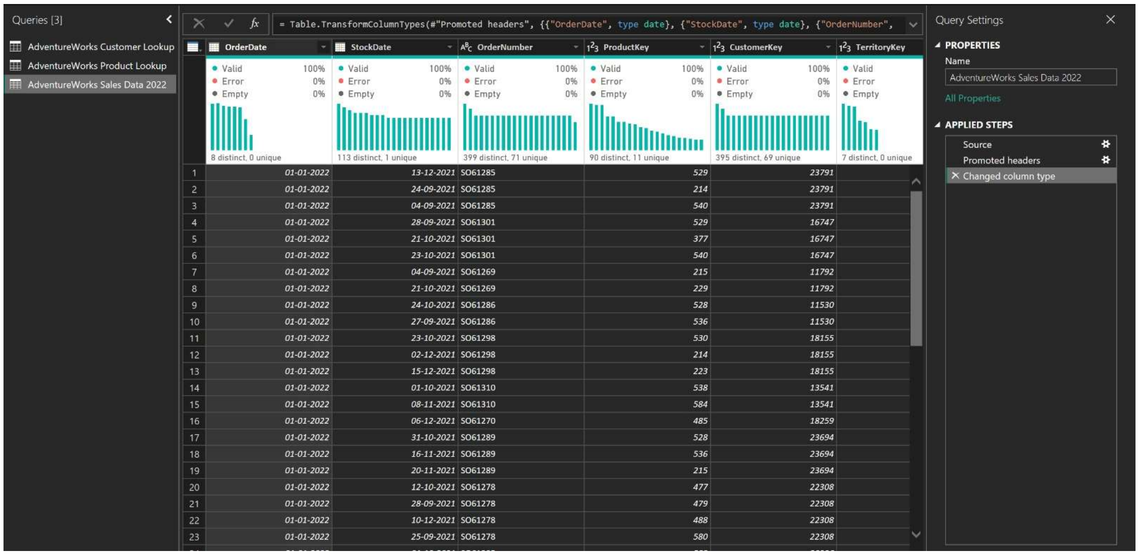

Figure 4(c): Sales data transformation in Power Query

➔ Data Modeling

Relationships were created between the fact table and the corresponding lookup tables using common key columns. One-to-many relationships were used where the lookup table contains unique keys and the sales table contains multiple corresponding records.

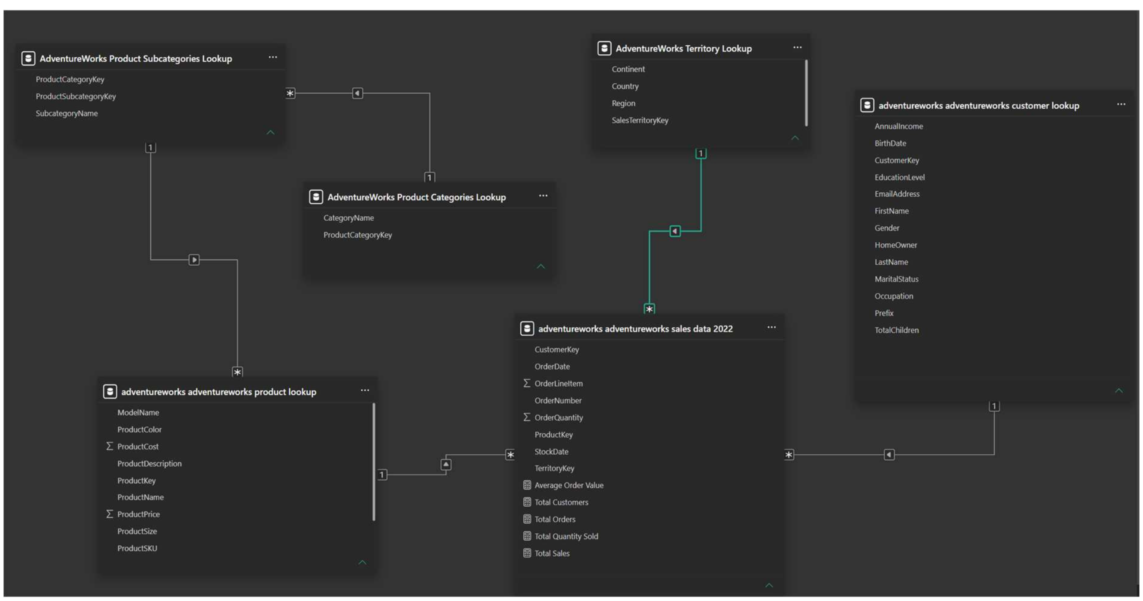

Figure 5: Data model and relationships in Power BI

DAX Measures Used

Total Sales

```
Total Sales =
SUMX(
    'adventureworks adventureworks sales data 2022',
    'adventureworks adventureworks sales data 2022'[OrderQuantity]
        * RELATED('adventureworks adventureworks product lookup'[ProductPrice])
)
```

Total Orders

```
Total Orders =
DISTINCTCOUNT(
    'adventureworks adventureworks sales data 2022'[OrderNumber]
)
```

Total Customers

```
Total Customers =
DISTINCTCOUNT(
    'adventureworks adventureworks sales data 2022'[CustomerKey]
)
```

Total Quantity Sold

```
Total Quantity Sold =
SUM(
    'adventureworks adventureworks sales data 2022'[OrderQuantity]
)
```

Average Order Value

```
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders],
    0
)
```

Dashboard Output

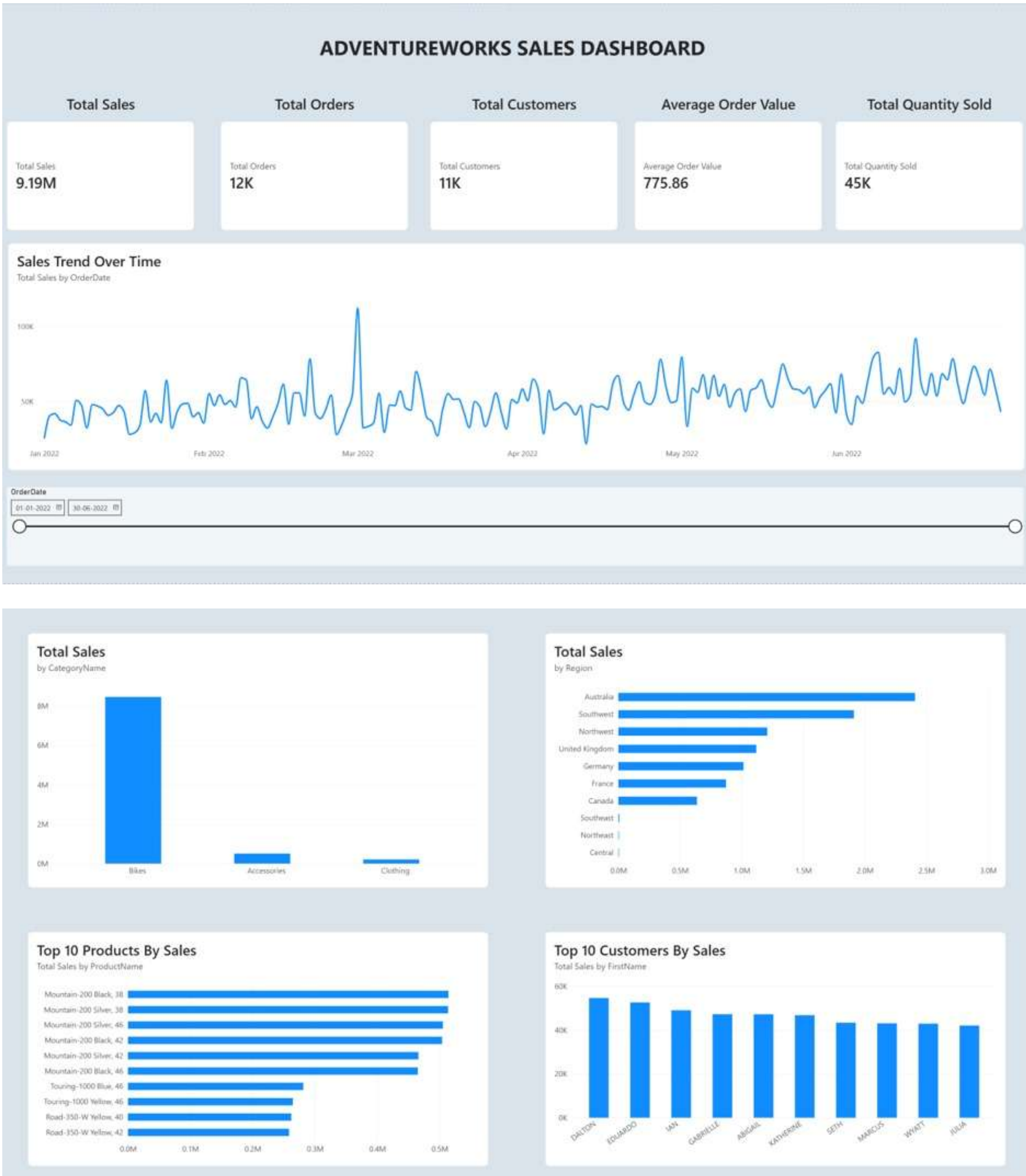

Figure 7: Final AdventureWorks Sales Dashboard

- **Total Sales Performance**
  The Total Sales KPI provides an overall view of the sales generated during the selected period.

- **Customer Analysis**
  Customer analysis helps identify customer-level sales performance and the contribution of individual customers to overall sales.

- **Product Analysis**
  The Top 10 Products by Sales visualization identifies products contributing the highest sales.

- **Regional Analysis**
  The Sales by Region visualization compares sales performance across different geographical regions.

- **Category Analysis**
  The Sales by Category visualization shows the contribution of different product categories to total sales.

- **Sales Trend**
  The Sales Trend Over Time visualization shows how sales vary across the selected dates.

Business Insights

Key Business Insights:

1. Total sales generated during the selected period were approximately ₹/currency 9.19M based on the dashboard.
2. The dashboard recorded approximately 12K orders.
3. Approximately 45K units were sold.
4. The Top 10 Products visualization highlights the products contributing most to sales.
5. Customer-level analysis identifies the highest-performing customers.
6. Sales trends vary across the 2022 period.
7. Regional and category analysis allows management to compare performance across different business segments.

Conclusion

The experiment successfully demonstrated the integration of AdventureWorks business datasets into Microsoft Power BI using MySQL and other data sources. The datasets were imported, transformed, and connected using appropriate relationships between sales, customer, product, category, subcategory, and territory tables.

DAX measures were developed to calculate important business KPIs such as Total Sales, Total Orders, Total Customers, Average Order Value, and Total Quantity Sold. An interactive dashboard was created to analyze product, customer, category, regional, and time-based sales performance.

The final dashboard provides a centralized view of business performance and enables users to interact with the data using filters and slicers.

This aligns with the reference's conclusion that the integrated data should be transformed into meaningful business insights and provide a centralized view for decision-making.

Learning Outcomes

Through this experiment, I learned:

- How to import business datasets from MySQL and other data sources into Power BI.
- How to use Power Query for data cleaning and transformation.
- How to create relationships between multiple tables.
- How to establish one-to-many relationships in a Power BI data model.
- How to develop DAX measures for business KPIs.
- How to create interactive dashboards using Power BI visuals.
- How to use slicers and filters for interactive analysis.
- How to analyze sales performance by product, customer, category, region, and time.
- How to generate meaningful business insights from raw datasets.